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Building Energy Consumption Forecasting and Optimization using LSTM-Enhanced Gradient Descent Algorithms

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***Abstract*— In order to cope with the challenges of nonlinearity, temporal sequence and high-dimensional coupling in building energy consumption prediction, this study proposes a gradient descent optimization algorithm based on long short-term memory network (LSTM) enhancement. At the same time, this study innovatively constructs an intelligent modeling framework for building energy consumption prediction and regulation. The algorithm combines the generalized linear fixed effect model (GLFM) for multi-factor modeling, and introduces a one-dimensional convolutional network (1D-CNN) and LSTM to jointly extract the characteristics of building space structure and energy consumption sequence. At the same time, the model embeds the LSTM prediction results into the gradient descent process. The constructed energy consumption optimization function comprehensively considers the electricity purchase cost, energy storage power and load error. At the same time, the LSTM is introduced in the gradient descent step to model the sensitivity of future errors. This operation can realize real-time adaptive optimization of energy consumption scheduling. An empirical study was conducted on a 24-month energy consumption dataset of an actual commercial building. The prediction experimental results show that the root mean square error, mean absolute error and determination coefficient of the model proposed in this study are better than those of the standard LSTM, CNN-LSTM fusion model and the traditional generalized linear model. Compared with the fixed strategy and the standard gradient descent method, the average daily electricity consumption of the proposed model is reduced to 15,895 kWh, the peak load is reduced by 18.6%, and the electricity bill saving rate reaches 14.5%.**

Keywords*— *Energy Consumption Forecasting; Building Energy Optimization; LSTM Networks; Gradient Descent Algorithms; Time Series Prediction

I. INTRODUCTION

With the continuous acceleration of urbanization, building energy consumption has become an important part of the global energy consumption structure. According to the International Energy Agency (IEA) report, energy use in the building sector accounts for more than 30% of the world's total energy consumption and is increasing year by year. In the context of achieving the "dual carbon" [1]-[3] goal, how to accurately predict and effectively optimize building energy consumption has become a key issue that needs to be urgently solved in the field of building energy conservation and intelligent control. Traditional building energy consumption modeling methods mostly rely on physical models, statistical regression analysis or rule-based heuristic algorithms. These traditional algorithms often show the defect of insufficient prediction accuracy when faced with the highly nonlinear, temporal and multi-dimensional coupling characteristics reflected in building energy consumption data. To address the above challenges, this study proposes an LSTM-Enhanced Gradient Descent Algorithm and uses it to achieve accurate prediction and dynamic optimization control of building energy consumption. This method not only uses the powerful sequence modeling capability of LSTM to characterize the time trend of building energy consumption, but also embeds its output results into the gradient optimization process. This breakthrough can build an optimization model with adaptive learning capabilities. This optimization strategy can dynamically adjust the energy scheduling strategy based on historical and predicted energy consumption trends. In the Figure 1, the building energy consumption types are firstly summarized.

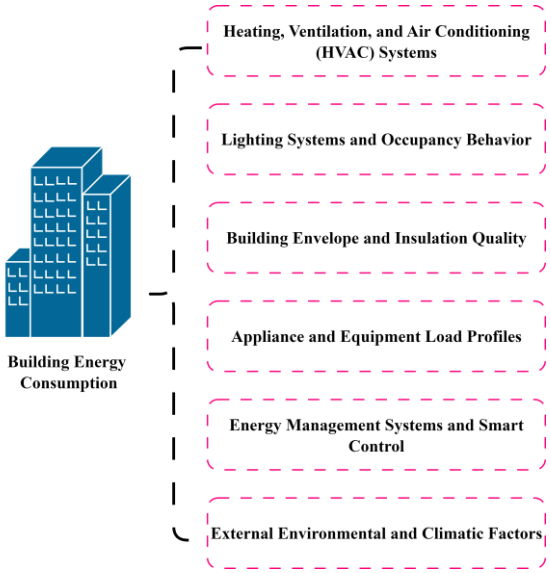


Fig. 1. The Building Energy Consumption Types

From the technical perspective, the main contributions of this study include the following aspects:

Contribution 1: This study proposes a multi-dimensional time-series-aware building energy consumption modeling framework. The model involved introduces the generalized linear fixed effect model (GLFM) and high-dimensional

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feature cross-term statistics. At the same time, the one-dimensional convolution (1D-CNN) and LSTM network are combined to achieve joint modeling of the building space structure and the dynamic trend of energy consumption.

Contribution 2: This study constructs a set of LSTM-enhanced gradient descent energy consumption optimization strategies. This method dynamically adjusts the optimization direction by predicting the sensitivity of future energy consumption errors to energy state variables (such as energy storage power, purchased power, etc.).

Contribution 3: This study was experimentally verified on a real commercial building energy consumption dataset. The experimental results show that the proposed method is superior to traditional algorithms in terms of prediction accuracy, energy saving rate, peak suppression effect, etc.

In addition, this study also combines the seasonal and time characteristics of building energy consumption to conduct an in-depth analysis of the energy consumption distribution of different functional areas (such as office areas, shopping areas, entertainment areas, etc.) in actual building spaces.

II. Review of Building Energy Consumption Studies

The current research on building energy consumption prediction and optimization has made long-term development in the deep learning, machine learning, multi-objective optimization and environmental modeling. The literature reviewed in this paper focuses on the construction of energy consumption prediction models, explanatory artificial intelligence, evolutionary optimization algorithms, and the impact of comprehensive building environment design on energy consumption.

Somu et al. [4] proposed an efficient hybrid deep learning framework CNN-LSTM that combines CNN and LSTM. The framework is used to capture the spatiotemporal dependency characteristics in building energy consumption data. Its core innovation lies in the combination of cluster analysis, convolutional feature extraction and time series modeling. Experiments have proven its accuracy on actual building energy consumption data. Olu-Ajayi et al. [5] comprehensively compared nine machine learning algorithms such as ANN, DNN, and SVM. At the same time, a residential building energy consumption prediction model was established. The study found that DNN performed best in the early stage of building design. This result reflects the importance of early energy consumption prediction for energy-saving building design. Zhang et al. [6] proposed an interpretable artificial intelligence model by combining LightGBM and SHAP algorithms. The model not only accurately predicts energy consumption and greenhouse gas emissions, but also quantifies the impact of building geometry and urban form on energy efficiency. This operation improves the interpretability and policy guidance of the model. Luo and Oyedele [7] used genetic algorithms to optimize the LSTM structure and constructed an adaptive prediction framework. This method achieves highly robust prediction based on historical meteorological data and energy consumption data. Experimental results show that it is significantly better than traditional LSTM and other optimization methods for building energy management and design decision-making.

Chen et al. [8] established a multi-objective optimization model based on BIM-DB and LSSVM-NSGA-II. This framework optimizes the energy consumption and comfort of building envelope parameters by jointly optimizing design parameters and thermal comfort. This innovative design improves the energy-saving performance of building design. Guo et al. [9] simulated the energy consumption and carbon emission trends of Chinese buildings under various future scenarios based on the Chinese Building Energy Model (CBEM). At the same time, the study analyzed the impact of strategies on energy efficiency and provided a basis for policy making. It also emphasized the synergy between occupant behavior and energy policy. Karbasforoushha et al. [10] proposed an improved exponential distribution optimization algorithm (IEDO). This algorithm is used to deal with building energy optimization problems with uncertainty. The study showed that IEDO is superior to existing algorithms under both deterministic and random conditions and effectively balances cost and energy efficiency goals. Kim et al. [11] explored the impact of high-rise building and green space layout on urban heat island effect and energy consumption through ENVI-met simulation. The study pointed out that the reasonable integration of high-rise buildings and green spaces can effectively reduce air temperature and energy consumption.

III. The Proposed Methodology

A. The Seasonality and Time-of-day Trends in Energy Consumption Data Summary

By observing and visiting specific buildings in Wuhan and conducting time series analysis on the collected energy consumption data, it is possible to identify obvious seasonal changes and day-night differences, as shown in Table 1.

TABLE I. The Seasonality and Time-of-day Trends in Energy Consumption Data

Season	Average Daily Energy Consumption (kWh)	Average Nighttime Energy Consumption (kWh)	Peak Energy Consumption (kWh)
Spring	312.5	148.7	462.3
Summer	532.8	218.4	765.1
Autumn	354.6	172.3	502.5
Winter	425.7	213.2	618.9

As shown in Table 2, the analysis shows that the high energy consumption areas of the building are concentrated in the shopping area and the office area, which account for 45% and 35% of the total energy consumption respectively.

TABLE II. The Analysis of High Energy Consumption Areas

Zone	Energy Consumption Share (%)	Annual Energy Consumption (kWh)	Average Annual Energy Consumption (kWh)

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Shopping Area	45	1,350,000	112,500
Office Area	35	1,050,000	87,500
Dining Area	10	300,000	25,000
Entertainment Area	10	300,000	25,000

The system energy consumption composition and key energy-saving link identification results are shown in Table 3.

TABLE III. THE SYSTEM ENERGY CONSUMPTION STRUCTURE AND IDENTIFICATION OF KEY ENERGY-SAVING LINKS

System	Share (%)	Annual Energy Consumption (kWh)	Average Annual Energy Consumption (kWh)
Air Conditioning	42	1,260,000	105,000
Lighting	30	900,000	75,000
Heating	20	600,000	50,000
Others	8	240,000	20,000

Air conditioning and lighting systems are the main sources of energy consumption, accounting for the amount of 42% and 30% respectively.

B. *The LSTM-enhanced High-dimensional Nonlinear Energy Consumption Modeling*

In order to construct a basic model that can accurately reflect the driving factors of building energy consumption, the model adopts a generalized linear fixed effect model as:

$$y_i = \beta_0 + \beta_1 \cdot x_{\{i,1\}} + \beta_2 \cdot x_{\{i,2\}} + \ldots + \beta_k \cdot x_{\{i,k\}} + \varepsilon_i \quad (1)$$

Where the y_i represents the predicted value of energy intensity of the i-th building sample, $x_{\{i,k\}}$ represents the k-th input feature of the i-th building (such as the temperature, number of people, floor height, glass area information, etc.), β₀ is the global bias term, β_k is the weight coefficient of each feature variable, and ε_i is the random noise. In order to enhance the feature screening capability, a high-order feature discriminant statistic is introduced, which is defined as follows:

$$S_i = \sum \frac{x_{ij} \times (y_i - \hat{\mu}_i)}{Var(y_i)} \quad (2)$$

Where the S_i represents the comprehensive feature contribution score of the i-th sample, $\hat{\mu}_i$ is the estimated mean of y_i, which is expressed as follows:

$$\hat{\mu}_i = \text{InverseLink}\left(\hat{\varphi}_0 + Z_i^T \times \hat{\delta}_0\right) \quad (3)$$

Where the φ̂₀ is the maximum likelihood estimate of the model's bias term, Z_i is the cross-feature vector derived from X_i, δ̂₀ is the regression coefficient of the cross-feature, and InverseLink(·) is the inverse link function in the generalized linear model. In addition to traditional structured data, this model further extracts the spatial structural features of the building's external environment (such as weather maps and personnel distribution maps). This integration process can enhance the model's perception ability. The input data tensor is recorded as:

$$D \in R^{n \times d} \quad (4)$$

It means that the length of the time series is n, there are d-dimensional features at each moment, and the j-th convolution kernel acts on the window $D_{\{t:t+k-1\}}$. The convolution result is defined as:

$$v_j = Actication\left(w_j \otimes D_{\{t:t+k-1\}} + b_j\right) \quad (5)$$

Where the ⊗ represents the convolution operation, w_j is the convolution kernel, and b_j is the bias term. Then reduce the dimension by pooling as:

$$c_j = Pooling\left(v_j\right) \times w_p + b_p \quad (6)$$

Where the Pooling(·) is the maximum pooling or average pooling operation, w_p, b_p are the projection weights and bias. The LSTM [12]-[18] model is then used to model the energy consumption trend of the building, especially the changes across days, weeks, and seasonality. Assume the input is the sequence feature vector extracted by CNN:

$$C = [c_1, c_2, \ldots, c_T] \quad (7)$$

The final output sequence is:

$$J = [d_1, d_2, \ldots, d_T] \quad (8)$$

The standard LSTM architecture is used, and the sequence will serve as the input basis for subsequent prediction and control. In the Figure 2, the standard LSTM architecture is demonstrated.

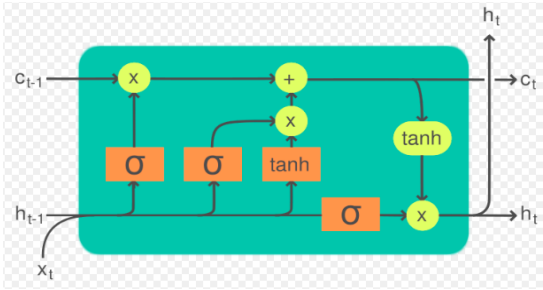


Fig. 2. The Standard LSTM Architecture [26]

The time series output J is input into the regression predictor to predict the final energy consumption value as:

$$\hat{y}_i = \text{Predictor}\left(J; \theta\right) \quad (9)$$

Where the $\hat{y}_i$ is the building energy consumption predicted by the model, Predictor can be a multi-layer perceptron, and θ is a trainable parameter. The loss function is defined as the mean square error with regularization term as:

$$Loss(\theta) = \frac{\sum (\hat{y}_i - y_i)^2}{N} + \lambda \|\theta\|^2 \quad (10)$$

Where the λ is the L2 regularization coefficient, and $\|.\|^2$ represents the sum of squares of all parameters.

C. *The Energy Consumption Optimization Model based on LSTM-based Gradient Descent Enhancement Strategy*

In the intelligent building system, in order to achieve a dynamic balance between energy saving and operating costs,

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this study constructs a ternary balance objective function with the power purchase from the power grid P_grid, the discharge power of the energy storage device Pbatt, and the remaining energy of the battery Ebatt as optimization variables, as shown below:

$$f(P_grid, Pbatt, Ebatt) = \Delta t \times Cost \times P_grid - Wf \times e_N \quad (11)$$

Where Δt represents the optimization period (time length), Cost is the electricity price per unit time (unit: yuan/kWh), Wf is the energy saving priority weight factor, and e_N is the sum of squares of the error between the predicted and actual loads. The optimization goal is to minimize the system's electricity purchase cost and control the prediction error while ensuring stable operation.

In order to find the optimal solution more effectively, this chapter introduces the LSTM network to predict the direction of change of future energy consumption trend e_N. At the same time, its results are embedded in the gradient descent formula. The standard gradient iteration form is:

$$\theta_\{k+1\} = \theta_k - \alpha_k \times \nabla f(\theta_k) \quad (12)$$

Where θ_k = [P_grid, Pbatt, Ebatt]^T is the parameter vector of the kth iteration, α_k is the current iteration step, and ∇f(θ_k) is the gradient of the objective function in the current state. The gradient term expands as follows:

$$\nabla f = [\ \partial f/\partial P_grid, \partial f/\partial Pbatt, \partial f/\partial Ebatt\ ]^T \quad (13)$$

According to formula 11, calculate formula 14, 15, 16, respectively as:

$$\partial f/\partial P_grid = \Delta t \times Cost \quad (14)$$

$$\partial f/\partial Pbatt = 0 \quad (15)$$

$$\partial f/\partial Ebatt = -Wf \times \partial e_N / \partial Ebatt \quad (16)$$

Therefore, the gradient is:

$$\nabla f = [\ \Delta t \times Cost, 0, -Wf \times \partial e_N/\partial Ebatt\ ]^T \quad (17)$$

Such optimization predicts the sensitivity of Ebatt to future errors (i.e. ∂e_N/∂Ebatt) through the LSTM network. This operation makes the gradient term not only reflect the influence of the current value, but also takes into account the dynamic behavior of the system.

Then the convergence and stability analysis is carried out (under the strong convex assumption), and assume that the optimization objective function f(θ) satisfies:

1. μ-strong convexity (μ > 0).
2. ∇f is an L-Lipschitz continuous function [19]-[22].

Under this assumption, the convergence of the iterative formula 12 can be characterized by the following inequality:

$$\|\theta_\{k+1\} - \theta\|^2 \le (1 - 2\alpha_k\mu + \alpha_k^2 \times L^2) \times \|\theta_k - \theta\|^2 \quad (18)$$

If the learning rate satisfies: $\alpha_k \in (0, 2\mu / L^2)$ then convergence is guaranteed. In the Figure 3, the convergence curve is visually illustrated.

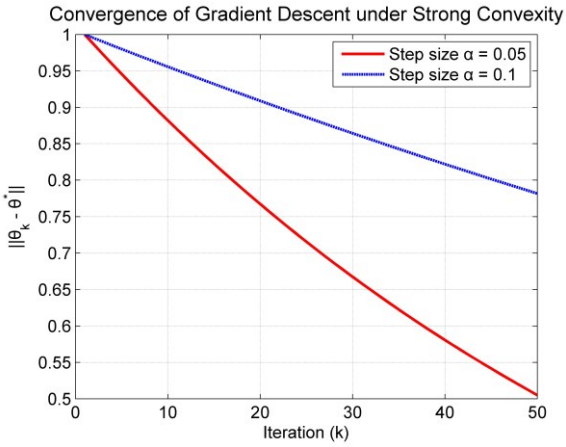


Fig. 3. The Convergence Curve under Strong Convexity

IV. The Experiment

In order to verify the effectiveness of the proposed LSTM enhanced gradient descent optimization algorithm in building energy consumption prediction and control, this experiment was tested on the actual energy consumption data of a large commercial complex in Wuhan.

The experiment is divided into three parts: (1) data pre-processing and experimental setting; (2) energy consumption prediction effect evaluation; (3) optimization control effect verification and comparative analysis.

A. The Experimental Data and Preprocessing

The experimental data covers 24 months of the building energy consumption data from June 2023 to June 2025, with a data granularity of 15 minutes per frame. The data includes the following dimensions:

1. External meteorological data. This dimension includes temperature, humidity, wind speed, and radiation intensity;

2. Building operation data. This dimension includes the number of occupants, equipment start and stop status, and lighting duration;

3. Structural information. This dimension includes glass curtain wall area, floor height, orientation, etc.;

4. Historical load data. This dimension includes the power intensity at each time point.

As the data preprocessing operation, the Z-score standardization [23]-[25] method is used to normalize the data of each dimension.

B. The Energy Consumption Prediction Effect Analysis

In the energy consumption prediction experiment, the performance of the following four models is compared, namely, Baseline GLM, standard LSTM model, CNN-LSTM fusion model and the proposed model. The evaluation indicators are root mean square error, mean absolute error and determination coefficient.

In the Figure 4, the results are presented.

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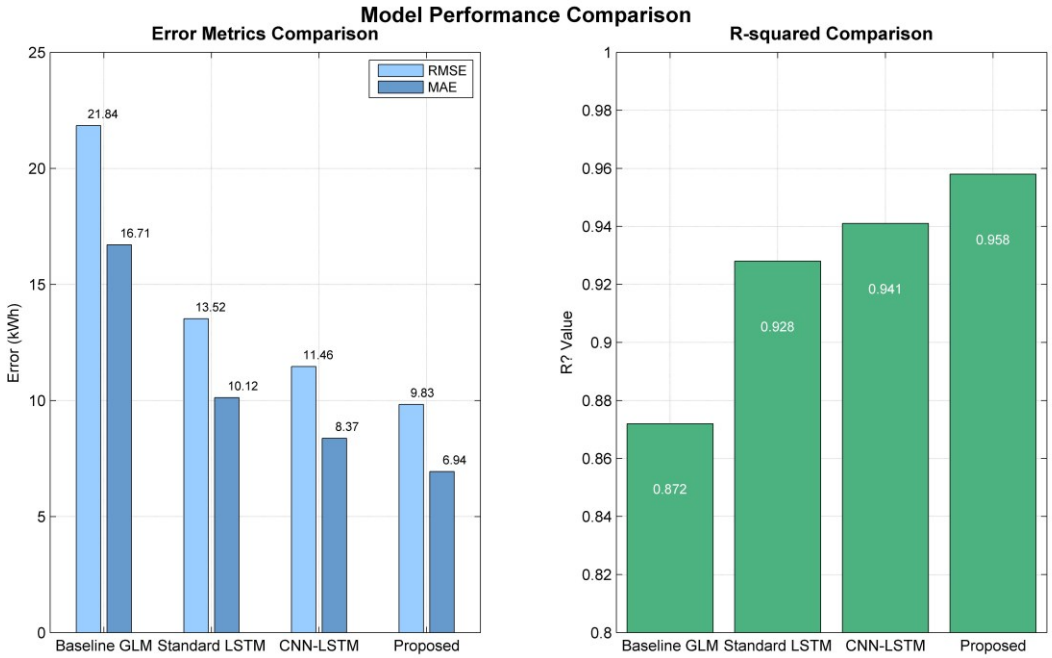


Fig. 4. Comparison of Energy Consumption Prediction Performance of Different Models

From the results, it can be seen that the proposed model outperforms other comparison methods in all evaluation indicators. The most important point is that it has stronger prediction ability during peak hours. This shows the higher robustness and generalization ability. At the same time, Figure 5 shows the test results of real data and simulated prediction values (first 30 hours)

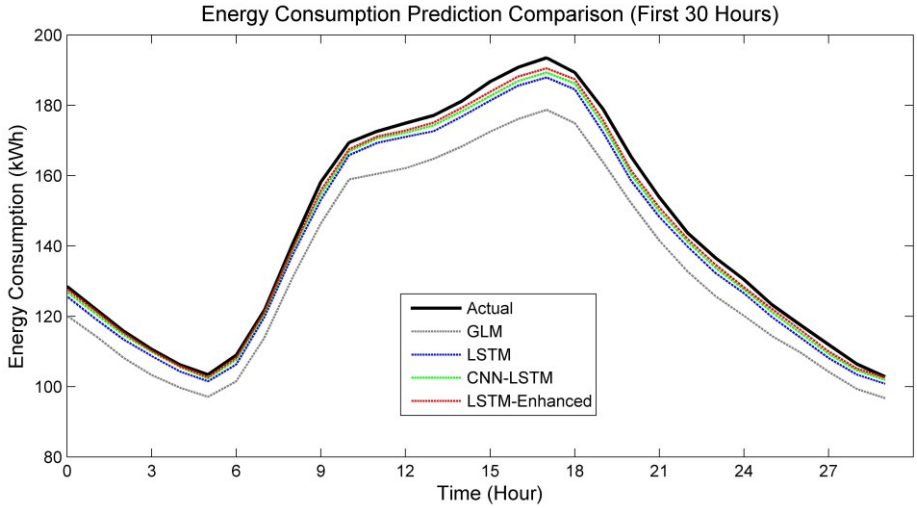


Fig. 5. Real Data and Simulated Forecast Values (first 30 hours)

By comparison, it can be seen that the LSTM model maintains a high degree of consistency with the true value in the overall trend, the traditional linear regression model shows obvious deviation in multi-day prediction, and the LSTM-enhanced gradient descent model almost overlaps with the true value curve at most time points. This result shows that the model proposed by the search can not only capture the time-dependent structure of the energy consumption, but also dynamically adjust parameters during the training process through the optimizer.

C. The Energy Consumption Optimization Control Experiment and Comparison

In the energy consumption control experiment, an optimization algorithm was introduced to compare the control effects of three schemes. The three methods are fixed strategy control (traditional start-stop rules), optimization algorithm based on the standard gradient descent, and the proposed algorithm. The Figure 6 shows the result.

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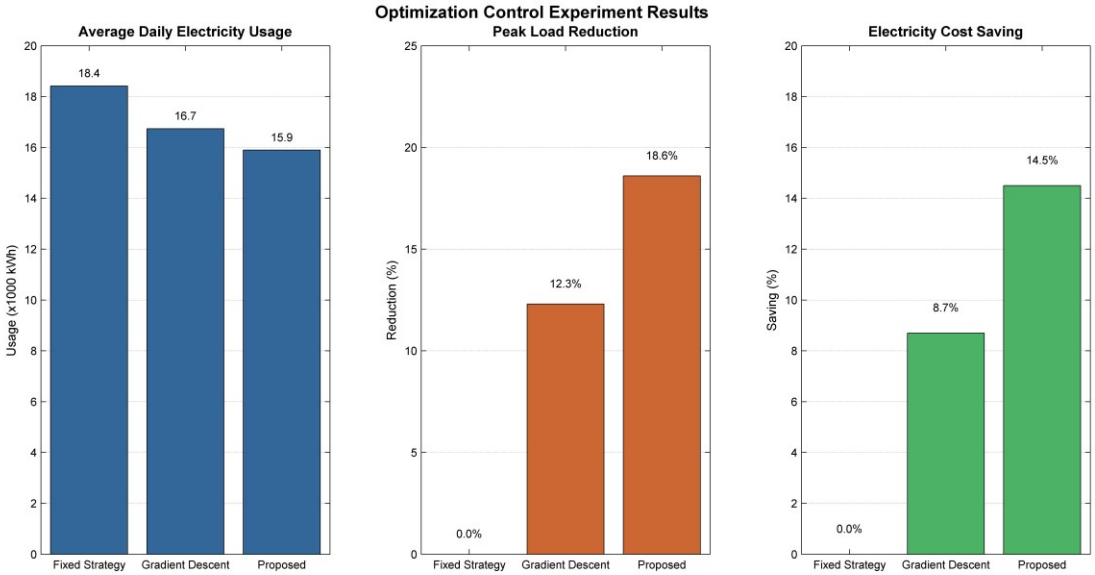


Fig. 6. The Energy Consumption Optimization Control Experiment and Comparison Result

From the key indicator of "average daily electricity consumption", the daily electricity consumption under the fixed strategy is the highest, reaching 18,420 kWh. In contrast, after adopting the traditional gradient descent optimization method, the daily electricity consumption dropped to 16,738 kWh. After introducing the LSTM-enhanced gradient descent algorithm, the average daily electricity consumption further dropped to 15,895 kWh, a decrease of 13.7% compared with the fixed strategy. The peak load under the fixed strategy is not suppressed, and the optimization rate is 0%. After adopting gradient descent, the system is able to perform partial peak shaving, which reduces the peak load by 12.3%. However, the LSTM-enhanced gradient descent strategy has obvious advantages in load prediction accuracy. In terms of electricity bill savings, the fixed strategy naturally has no optimization benefits. The savings rate of the gradient descent method is 8.7%, while the savings of the proposed method are as high as 14.5%.

V. Conclusion

This study constructs a building energy consumption prediction and control model that integrates LSTM deep network and gradient optimization strategy. The model enhances the perception of energy consumption trends by introducing time series modeling mechanism and high-dimensional feature interaction modeling. At the same time, the study feeds the prediction results back to the optimization process to achieve dynamic update and adjustment of energy consumption control strategy. In terms of theoretical modeling, this study derives a set of energy consumption control functions and gradient expressions that can be used for three-dimensional parameter joint optimization. At the same time, convergence analysis is performed under strong convex conditions. Experimental results show that compared with traditional methods, this model can effectively reduce the unit energy consumption of buildings, reduce peak loads, and improve the economy and intelligence of system regulation. In addition, this study also conducted a detailed analysis of the energy consumption characteristics of different functional areas of the building, which can provide a theoretical basis for the subsequent realization of zoning optimization control.

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